SUSTAINABLE PRACTICES & CSR THROUGH MULTIDISCIPLINARY LENS:

A PATHWAY TO RURAL DEVELOPMENT

Student Innovation Power Lab The Student Innovation Power Lab is a Multidisciplinary Student Group of Entry Level Students in Science and Technology. The focus is on practical learning through real-life problem-solving case studies.

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Course: 3-Year Diploma in Electrical Engineering

Institution: JIS College of Engineering ; Council: WBSCTVESD

(84% in 1st Sem; Appeared 2nd Sem)

HS (PCBM & English)

TITLE

- Controller Algorithm and Semiconductor Switching for Optimization of Renewable Energy in Micro Power System for Home Appliances with Scalability to EV Modules for Sustainable Energy Development.

Sub Theme (CSR Alignment)

1. CSR driven Training and Capacity Building
2. Sustainable Livelihoods through Allied Activities in Rural Development
3. Any Other Areas Related to the Theme - Optimization of Renewable Energy in Micro Power System for Home Appliances for Sustainable Energy Development (Academic R&D only)

NEED OF THE STUDY

This study presents the concept of integration of Controller Management System with Semiconductor Switching in Micro Power System for Home Appliances. The approach emphasizes safe and sustainable Smart dual-input energy management concepts for Optimization of Renewable Energy as an easy, affordable alternative solution built upon state-of-the-art efficiency.

BRIEF LITERATURE

The project is based on established research in:

- Niasar & Binder (2021): Micro-power management and distributed energy optimization.
- Rashid (2017): Semiconductor switching and power electronics.
- Tata Power DDL: Smart-grid and dual-input energy management concepts adapted to a small-scale system. These references provide the theoretical foundation for the controller algorithm and semiconductor switching design.

OBJECTIVES

- Develop a compact, portable, and low-cost renewable energy micro power system.
- Optimize renewable energy using controller algorithms and semiconductor switching.
- Ensure user safety by limiting operation to ≤50 V DC.
- Support essential household appliances during power cuts.
- Build practical engineering skills among students and promote rural entrepreneurship through CSR-driven training. and explores pathways for Sustainable Livelihoods through skill development in allied renewable energy activities in rural areas.

METHODOLOGY

The project follows engineering approach:

- The Controller Unit is the Brain of the System built upon with automatically sensing logic (from multi volt capped up to DC 50 V ≤) and Manages Power Flow to charge stable DC voltage (minimum 12 V) and to run device load with continuous charging and discharging feature.
- Using BIS-certified Li-ion (300 – 500 cycle) or LiFePO4 (5000 cycle use) ensures low cost and higher efficiency (C rate).
- Device-wise selection of silicon panel architecture - Flexible Mono PERC/ Small size / Portable Foldable/Nano with Common Controlling Unit with sustainable Smart dual-input logic uses Grid & Renewable dual charging feature ensures uninterrupted power in any weather conditions.
- Field testing to evaluate real-world performance.

FINDINGS

The research successfully developed and demonstrated five working prototype modules:

- Nano Solar Power Bank
- LED Extender & Flood Light
- Solar BLDC Fan
- Mini Power Hub (384 Wh)
- Decorative Electro-Craft
- ANNEXURE 'A' **CLICK TO VIEW**
- PRODUCT IMAGE WITH CONFIGURATION DETAILS **CLICK TO VIEW**

Watch Video Click to view (Electro Crafts on Vase (glass/ceramic/real plant etc Lights and LED Extender, Power bank

Note: This link below contains video with movie song which was used previously in a corporate presentation in HPCL Bio Fuel, Patna(maybe run at the discretion of the organiser) - Click to view

CONCLUSION

The study concludes that a low-cost, extra-low-voltage micro power system can safely meet essential domestic energy needs in underserved communities. It also proves the effectiveness of the "learning by developing" approach for student training and supports sustainable rural development through renewable energy and skill-building.

FURTHER STUDY

Future research may focus on:

- Scaling the system for Electric Vehicle (EV) applications.
- Developing EV Solar Platforms (EVSP).
- Improving controller algorithms and semiconductor switching efficiency.
- Enhancing battery management, thermal management, and inverter performance.
- Expanding vocational training and real-world field deployment for rural communities.

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